

Assignment-02

Ca-n

A Strong real-world example that combines Parallel computing and Networked System is online Multiplayer gaming, Particularly in Large-scale games like Fortnite, Call of Duty, MMORPGs like World of Warcraft.

Application: online Multiplayer Gaming

Use of Parallel Computing:

→ Game, Servers and client devices (PCs, Consoles) use Parallel Computing to handle multiple tasks at the same time:

→ Rendering Complex 3D graphics using multi-core GPUs and CPUs

→ Simultaneously managing game Physics, animations, AI behaviors and Player input.

→ Processing thousands of concurrent game events efficiently.

Use of Networked System:

→ These games rely on networked system to:

- Enable real-time communication between players across the globe.

- Sync game states between client and servers to maintain a consistent experience.

- Allow cloud-based matchmaking, Player authentication and game state storage.

Why These Technologies are important:

- Performance & Real-Time Interaction: Without Parallel Computing, the heavy computation load (graphics, logic, AI) would lead to lag and reduced frame rates. Without networked systems,

Multiplayer interaction wouldn't be possible.

- Scalability Games Can Support millions of players simultaneously across different regions.
- Reliability & speed: parallel processing helps in faster load times and seamless transitions; networking ensures timely synchronization and event delivery with minimal latency.